1.7 Runge's Theorem. for compact subsets of open sets

Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Let $E \subset \mathbb{C}_\infty - K$ be a subset that meets each component of $\mathbb{C}_\infty - K$.

Then for any **holomorphic function** $f: G \to \mathbb{C}$ on G and any $\epsilon > 0$,

there exists a rational function $R(z)$ whose poles lie only in E

such that $|f(z) - R(z)| < \epsilon$ for all $z \in K$

(R approximates f; "uniformly on K")

Compare:

1.7 Runge's Theorem. *Let K be a compact subset of $\mathbb{C}$ and let E be a subset of $\mathbb{C}_\infty - K$ that meets each component of $\mathbb{C}_\infty - K$. If f is analytic in an open set containing K and $\epsilon > 0$ then there is a rational function $R(z)$ whose only poles lie in E and such that*

$$|f(z) - R(z)| < \epsilon$$

for all z in K.

Note: Regard each bounded component of $\mathbb{C}_\infty - K$ as a "hole" of K

The condition E meets each component of $\mathbb{C}_\infty - K$ means that

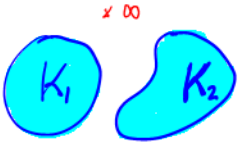
- E has a point in each "hole" of K

and

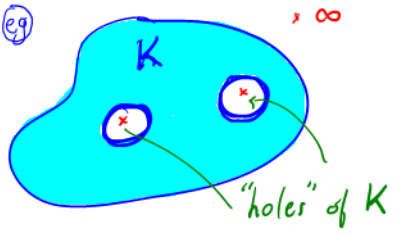
- E has a point near ∞ in $\mathbb{C}_\infty$ (ie. in the unique unbounded component of $\mathbb{C}_\infty - K$)

(usually just take ∞ itself)

eg



eg



Note: By taking $\epsilon = 1/n$, we get a sequence of rational functions whose poles lie only in E which converges normally on K to f

Example: G = open disk $B(a; R)$, $E = \{\infty\}$

power series expansion $f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n$

converges normally on compact disks $K \subset G$

Example: G = open annulus $\mathrm{ann}(a; R_1, R_2)$, $E = \{a, \infty\}$

Laurent series expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n (z-a)^n$

converges normally on compact annuli $K \subset G$

Proposition. (Weak Runge's Theorem — case of $E = \mathbb{C}_\infty - K$) ← largest possible

Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Then for any holomorphic function $f: G \to \mathbb{C}$ on G and any $\epsilon > 0$,

there exists a rational function $R(z)$ whose poles lie in $\mathbb{C}_\infty - K$

such that $|f(z) - R(z)| < \epsilon$ for all $z \in K$ uniformly on K

Proof. Recall :

Proposition. (Separation Lemma)

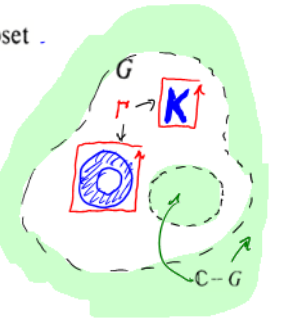
Let $G \subset \mathbb{C}$ be an open set, and let $K \subset G$ be a compact subset.

Then there exists a 1-cycle Γ in $G - K \subset G$ such that

$$n(\Gamma; w) = 0 \quad \text{for all } w \in \mathbb{C} - G$$

$$\text{and} \quad n(\Gamma; z) = 1 \quad \text{for all } z \in K$$

Moreover, Γ consists of a finite number of closed polygons.



(Enhanced Separation Lemma)

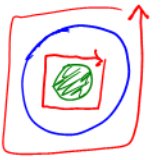
compare

Chapter VIII

1.1 Proposition. *Let K be a compact subset of the region G; then there are straight line segments $\gamma_1, \ldots, \gamma_n$ in $G - K$ such that for every function f in $H(G)$,*

$$f(z) = \sum_{k=1}^{n} \frac{1}{2\pi i} \int_{\gamma_k} \frac{f(w)}{w - z}\, dw$$

for all z in K. The line segments form a finite number of closed polygons.



- First, apply the Enhanced Separation Lemma (= Separation Lemma + global Cauchy's integral formula) to get a 1-cycle Γ in $G - K \subset G$ consisting of a finite number of closed polygons such that

$$f(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(w)}{w - z}\, dw \quad \text{for all } z \in K$$

- Write $\Gamma = E_1 + \cdots + E_n$ as a finite sum of straight line segments E_j in $G - K$

Then $$f(z) = \sum_{j=1}^{n} \underbrace{\frac{1}{2\pi i} \int_{E_j} \frac{f(w)}{w - z}\, dw}_{=: f_j} \quad \text{for all } z \in K$$

It suffices to show the approximation claim for each f_j

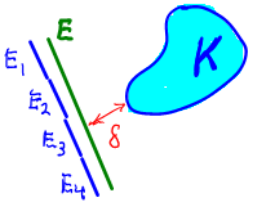
i.e. without loss of generality, we may assume $f: G \to \mathbb{C}$ is given by

$$f(z) = \frac{1}{2\pi i} \int_{E} \frac{\varphi(w)}{w - z}\, dw$$

where $\varphi \left(= f\big|_E\right)$ is a continuous function on a straight line segment E in $G - K$

- Let $\delta := d(K, E) > 0$

Write $E = E_1 + \cdots + E_n$ where each E_j is of length $< \delta$



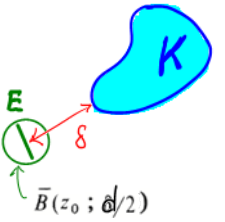
It suffices to show the approximation claim for each E_j

i.e. without loss of generality, we may assume further that

E is of length $d < \delta = d(K, E)$

- Let $z_0 \in E$ be the midpoint of E

Then the closed disks $\bar{B}(z_0 ; d/2)$ having E as diameter is disjoint from K



- The function $f(z) = \dfrac{1}{2\pi i} \displaystyle\int_E \frac{\varphi(w)}{w - z}\, dw$ is holomorphic on (int. rep. fn)

$\mathbb{C} - E$
$\cup$
$\mathbb{C} - \bar{B}(z_0 ; d/2)$
$= \operatorname{ann}(z_0 ; d/2, \infty)$

Consider its Laurent series expansion at z_0

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$ converging locally normally on $\operatorname{ann}(z_0 ; \delta/2, \infty)$

Recall: * **Theorem.** *Let* $a \in \mathbb{C}$ and $0 \le R_1 < R_2 \le \infty$ and $G = \operatorname{ann}(a; R_1, R_2)$ *and let* $f: G \to \mathbb{C}$ *be a holomorphic function on the annulus* G

(a) **Laurent Decomposition**

There exist holomorphic functions ① $f_a : B(a; R_2) \to \mathbb{C}$ and ② $f_\infty : \operatorname{ann}(a; R_1, \infty) \to \mathbb{C}$ with ③ $\lim_{z \to \infty} f_\infty(z) = 0$ in $\mathbb{C}$

such that ④ $f = f_a + f_\infty$ as holomorphic functions on $G = B(a; R_2) \cap \operatorname{ann}(a; R_1, \infty)$

Moreover, f_a and f_∞ with the above properties are uniquely determined by f

(b) **Laurent Series Expansion**

The function f is representable by a Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - a)^n$$ converging locally normally on G (annulus)

For any integer $n \in \mathbb{Z}$, *the coefficient* a_n is given by

$$a_n = \frac{1}{2\pi i} \int_{C(a; r)} \frac{f(w)}{(w - a)^{n+1}}\, dw \quad \text{in } \mathbb{C}$$ for any $R_1 < r < R_2$.

Its Laurent decomposition is thus

$$f(z) = \underbrace{\sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}}_{f_\infty(z)} + \underbrace{\sum_{n=0}^{\infty} a_n (z-z_0)^n}_{f_{z_0}(z)}$$

with f_{z_0} holomorphic on $B(z_0;\infty) = \mathbb{C}$ ie. an entire function

and f_∞ holomorphic on $\mathrm{ann}(z_0;\delta/2,\infty)$ and $\lim_{z\to\infty} f_\infty(z) = 0$

- Since $f(z) = \frac{1}{2\pi i}\int_E \frac{\varphi(w)}{w-z}\,dw$ clearly satisfies $\lim_{z\to\infty} f(z) = 0$,

it follows that $\lim_{z\to\infty} f_{z_0}(z) = 0$ as well, and hence $f_{z_0} = 0$

by Liouville's Theorem

Hence, $f(z) = \sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}$ converging locally normally on $\mathrm{ann}(z_0;\delta/2\ \infty)$

- Since K is disjoint from $\bar{B}(z_0;\delta/2)$,

we have $K \subset \mathbb{C} - \bar{B}(z_0;\delta/2) = \mathrm{ann}(z_0;\delta/2,\infty)$,

so $f(z) = \sum_{n=1}^{\infty} a_{-n}(z-z_0)^{-n}$ converges normally on compact K

Given any $\epsilon>0$, a sufficiently large $N \in \mathbb{N}$ then gives

$$\sup_{z\in K}\left| f(z) - \sum_{n=1}^{N} a_{-n}(z-z_0)^{-n}\right| < \epsilon$$

so $R(z) = \sum_{n=1}^{N} a_{-n}(z-z_0)^{-n}$ is a rational function $R(z)$

whose pole z_0 lies in $\mathbb{C}_\infty - K$

giving the desired approximation to f on K ##